

Paper proposal: System and energy dependence of near-side jet-like correlations

- PAs: Christine Nattrass, Jana Bielcikova, Jörn Putschke
- Study of near-side Jet and Ridge components of h-h correlations in Cu+Cu, Au+Au at $\sqrt{s_{NN}} = 62$ GeV, 200 GeV
- 8 figures proposed
- Beginning of analysis note web page:
<http://rhig.physics.yale.edu/~nattrass/Research/web/index.html>
- Target journal: PRC
- Planned as rapid communication (5 pages)?

System and energy dependence of near-side jet-like correlations

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(Dated: July 28, 2008)

Target Journal:PRC

Two-particle high- p_T triggered correlations in $\Delta\phi$ and $\Delta\eta$ in $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV in STAR at RHIC are presented. The N_{part} , $p_T^{associated}$, and $p_T^{trigger}$ dependence of the near-side yield per trigger of the component of the near-side at $\Delta\eta \approx 0$ and $\Delta\phi \approx 0$ is compared to the yield in $d + Au$ and to Pythia. While there are slight deviations, the behavior is remarkably consistent for the different collision systems and consistent up to a scaling factor with Pythia. This suggests that the structure at $\Delta\eta \approx 0$ and $\Delta\phi \approx 0$ is produced dominantly fragmentation. Pythia calculations suggest that for the same $p_T^{trigger}$ different jet energies are selected, providing a tool to study the production of the correlations at large $\Delta\eta$, the *Ridge*, at different jet energies. The *Ridge* is found to be substantially smaller in collisions at $\sqrt{s_{NN}} = 62$ GeV than in collisions at $\sqrt{s_{NN}} = 200$ GeV at the same N_{part} . This is potentially a powerful measurement to test models used to describe production mechanisms for the *Ridge*.

PACS numbers: 25.75.-q, 21.65.Qr, 24.85.+p, 25.75.Bh

Figure 1

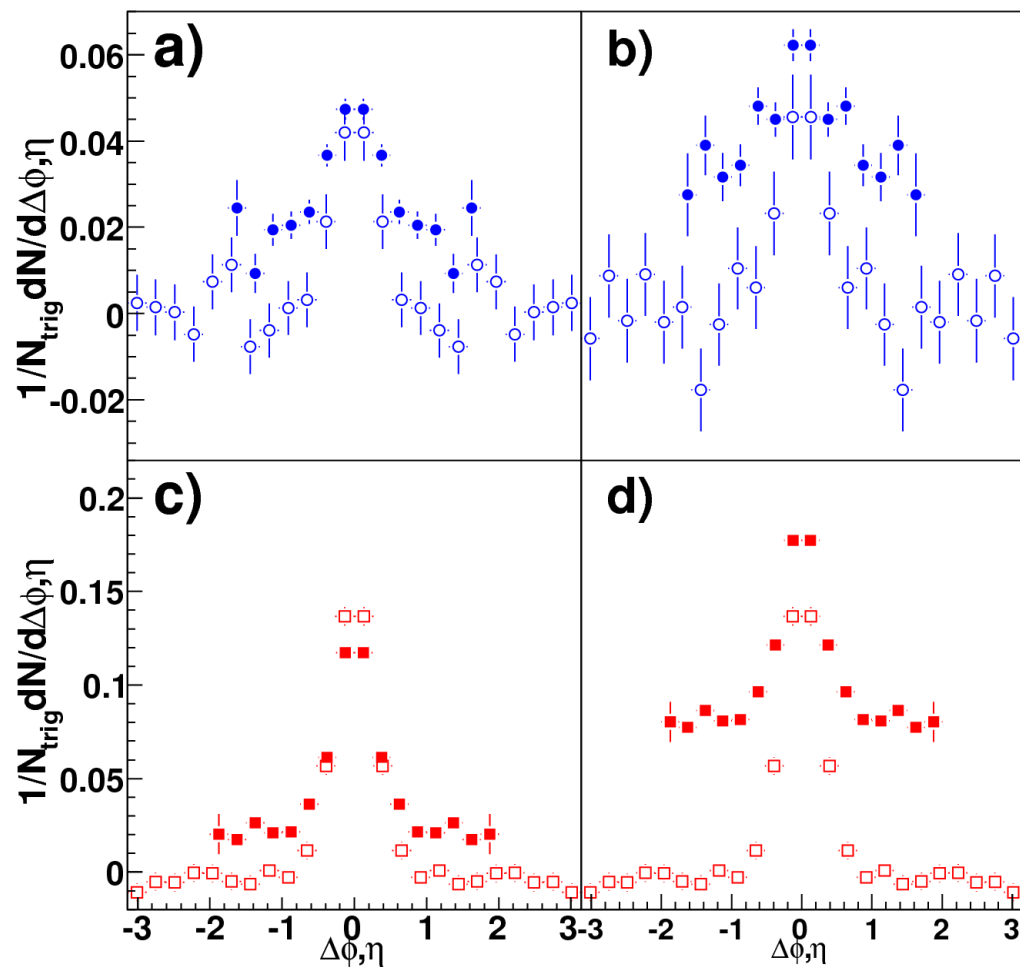


FIG. 1: Sample correlations for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$ for 0-10% centrality for (a) $Cu + Cu$ at $\sqrt{s_{NN}} = 62$ GeV (b) $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV (c) $Cu + Cu$ at $\sqrt{s_{NN}} = 200$ GeV and (d) $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV. The dependence of the *Jet* is shown as a function of $\Delta\phi$ in open symbols and $\Delta\eta$ in closed symbols. Color online.

Figure 2

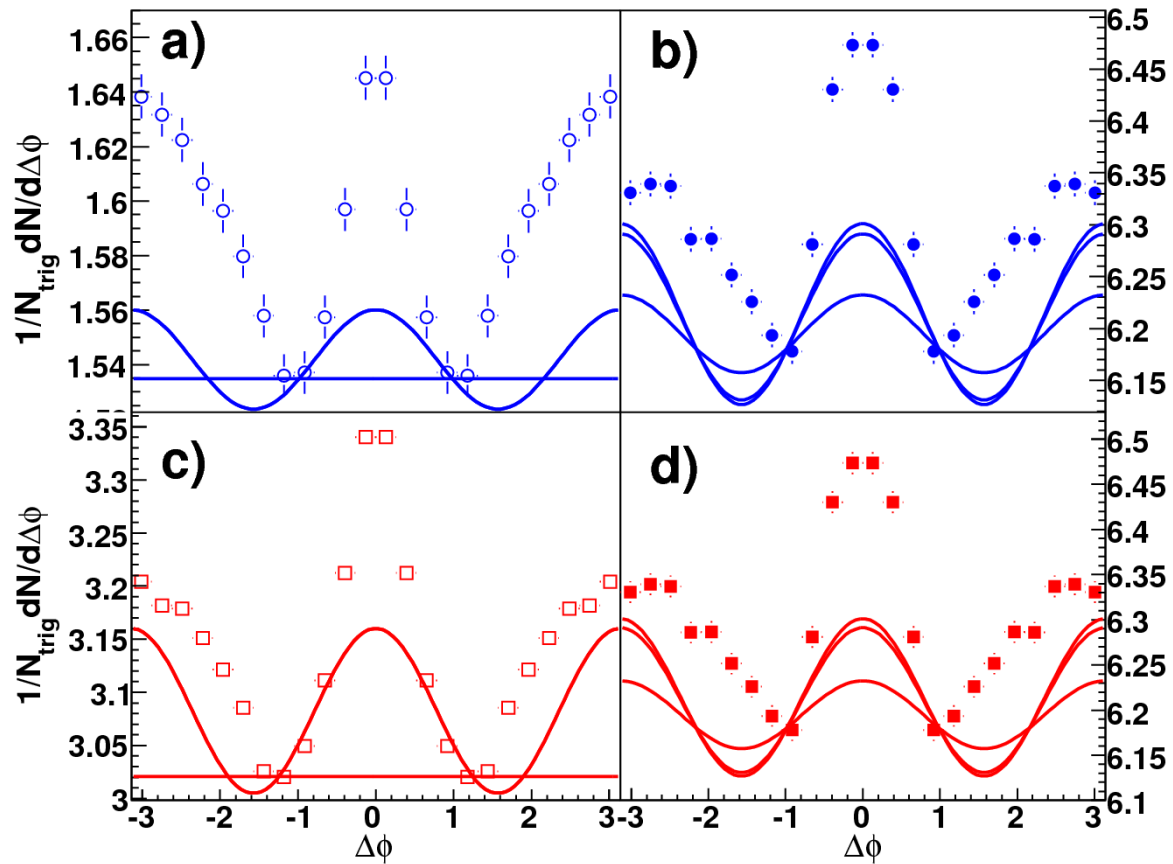


FIG. 2: Sample correlations for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$ for 0-10% centrality for (a) $Cu + Cu$ at $\sqrt{s_{NN}} = 62$ GeV (b) $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV (c) $Cu + Cu$ at $\sqrt{s_{NN}} = 200$ GeV and (d) $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV. Lines show the background using the ZYAM method with the range of v_2 used for the determination of the systematic errors.

Figure 3

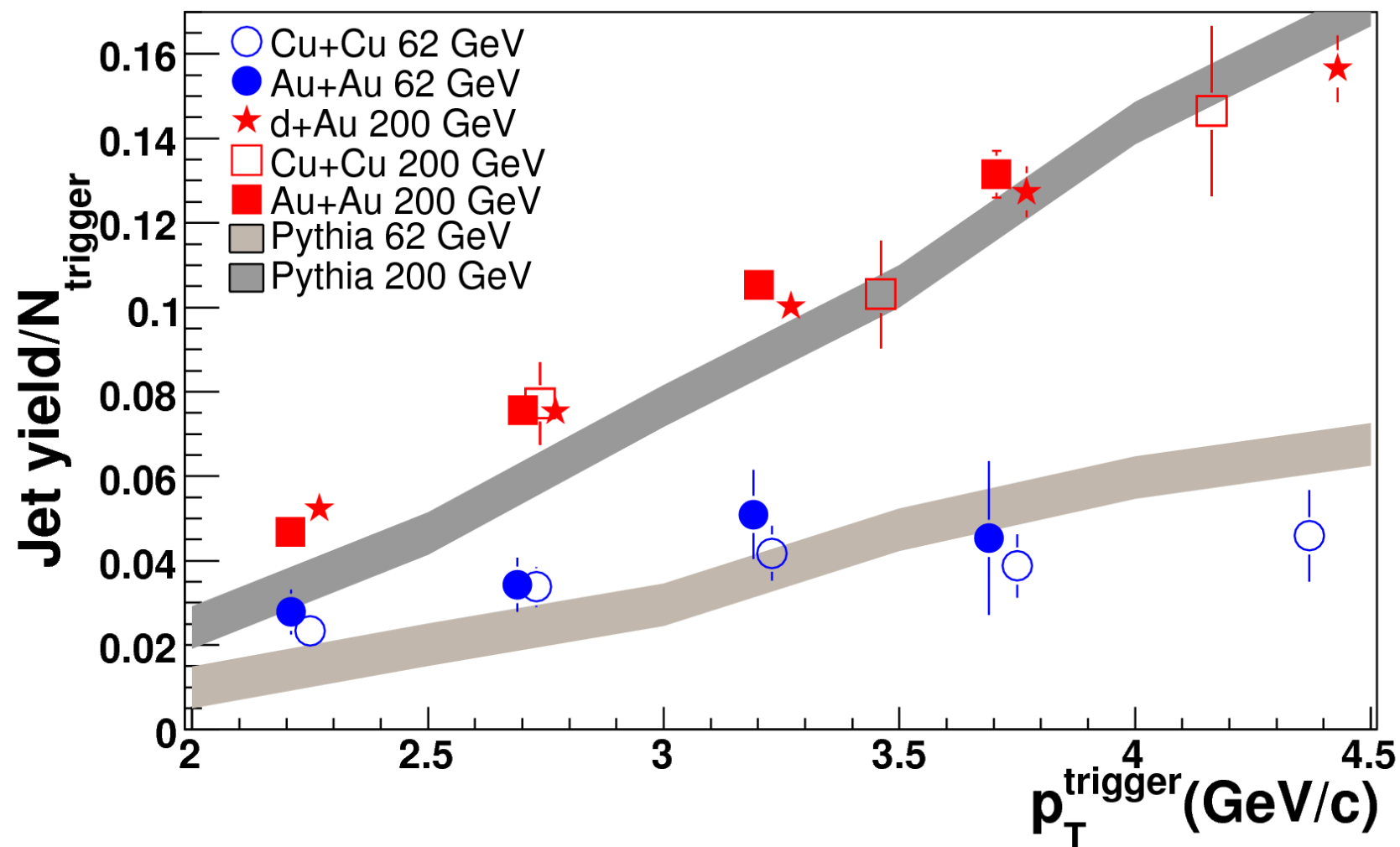


FIG. 3: Dependence of Jet yield on $p_T^{trigger}$. The yield is nearly independent of collision system and is consistent with the $d + Au$. The shape of the dependence on $p_T^{trigger}$ is well described by Pythia. Color online.

Figure 4

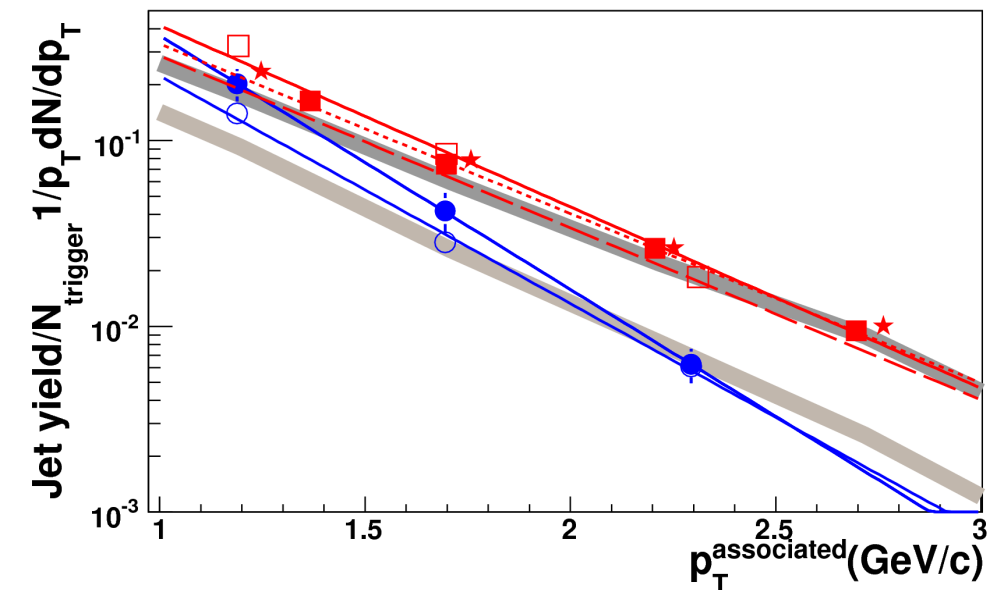


FIG. 4: Dependence of *Jet* yield on $p_T^{associated}$. The key is the same as in figure 3. The yield is nearly independent of collision system and is consistent with the $d + Au$. The spectrum from Pythia is slightly harder, which may explain why the $p_T^{trigger}$ dependence must be scaled. Color online.

TABLE I: Inverse slope parameter k (MeV/c) of $p_T^{associated}$ for fits of data in Figure 4. The inverse slope parameter from a fit to π^- in $Au + Au$ from [3] above 1.0 GeV/c is $k = 280.9 \pm 0.4$ MeV/c for $\sqrt{s_{NN}} = 62$ GeV and is $k = 330.9 \pm 0.3$ MeV/c for $\sqrt{s_{NN}} = 200$ GeV. Statistical errors only.

	$\sqrt{s_{NN}} = 62$ GeV	$\sqrt{s_{NN}} = 200$ GeV
	h-h	h-h
$Au + Au$ Jet	317 ± 26	478 ± 8
$Cu + Cu$ Jet	355 ± 21	445 ± 20
$d + Au$ Jet		469 ± 8
Pythia	417 ± 9	491 ± 3

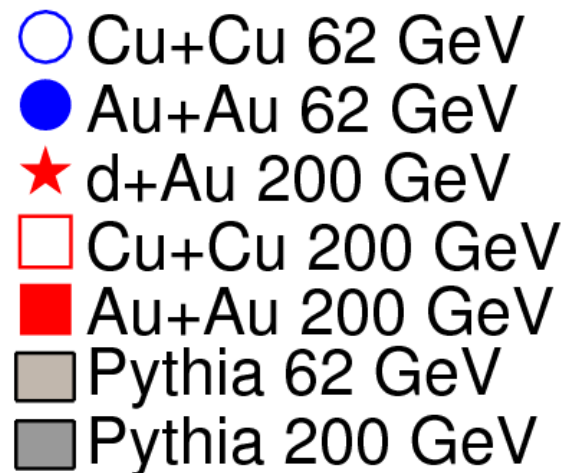


Figure 5

- Cu+Cu 62 GeV
- Au+Au 62 GeV
- ★ d+Au 200 GeV
- Cu+Cu 200 GeV
- Au+Au 200 GeV
- Pythia 62 GeV
- Pythia 200 GeV

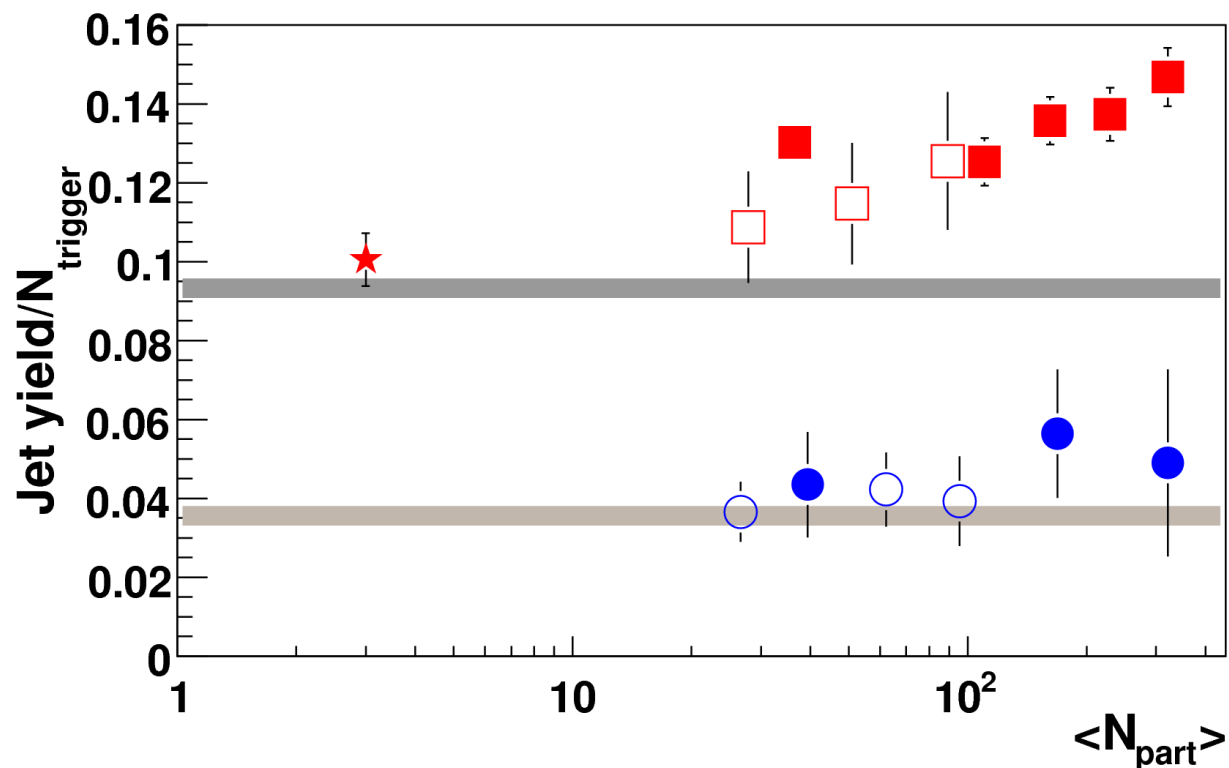


FIG. 5: Dependence of *Jet* yield on N_{part} . The *Jet* yield in $Cu + Cu$ and $Au + Au$ are within error at both $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV. There is a slight increase with N_{part} . This may be attributed to the method. If the *Ridge* is not completely independent of $\Delta\eta$ it would contribute more to the yield at high N_{part} . The yield at $\sqrt{s_{NN}} = 62$ GeV is about 40% lower than at $\sqrt{s_{NN}} = 200$ GeV. This would be expected if the *Jet* were primarily from fragmentation.

Color online.

Figure 6

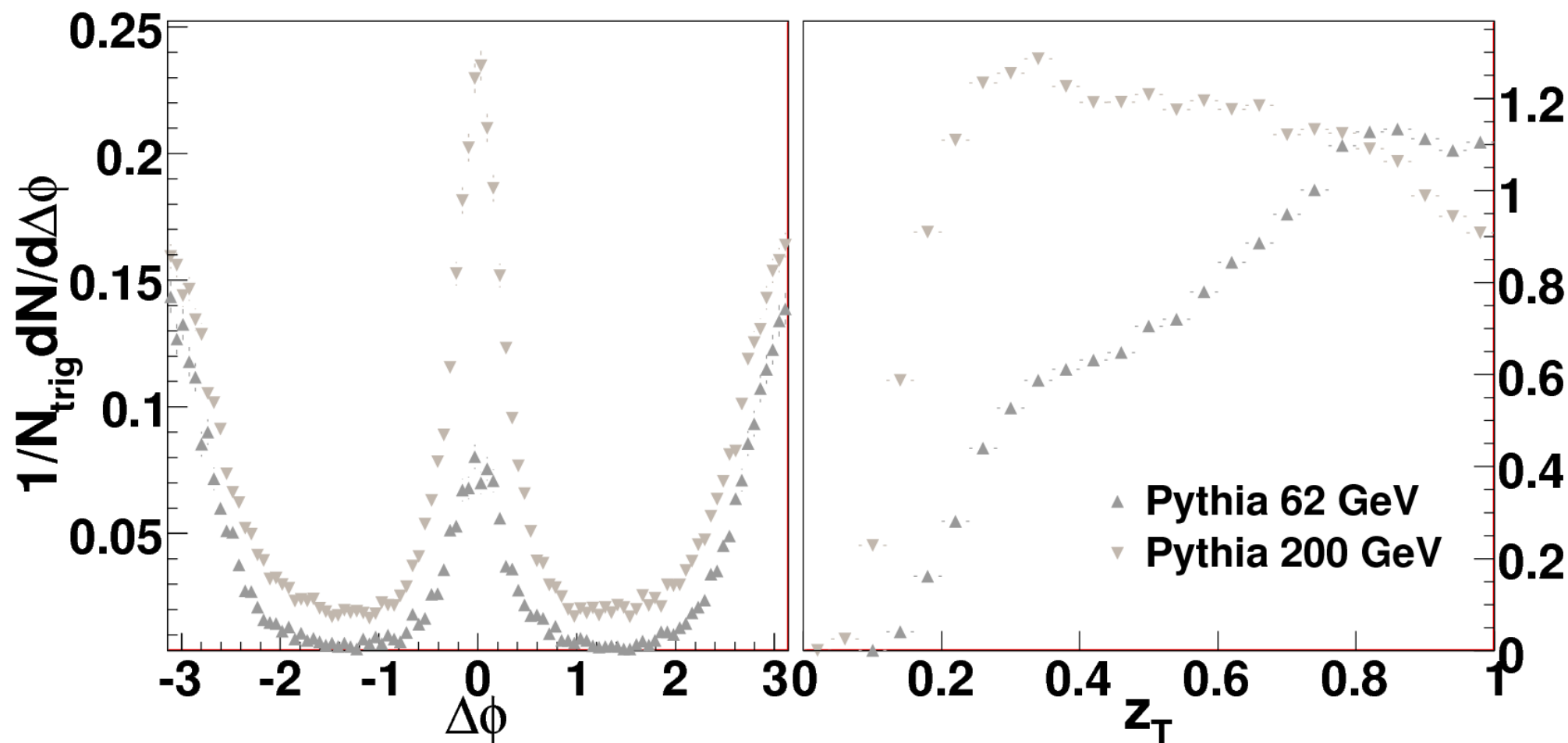


FIG. 6: z_T distribution at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV as predicted by Pythia for $3.0 < p_T^{\text{trigger}} < 6.0$ GeV/c and $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$.

Figure 7

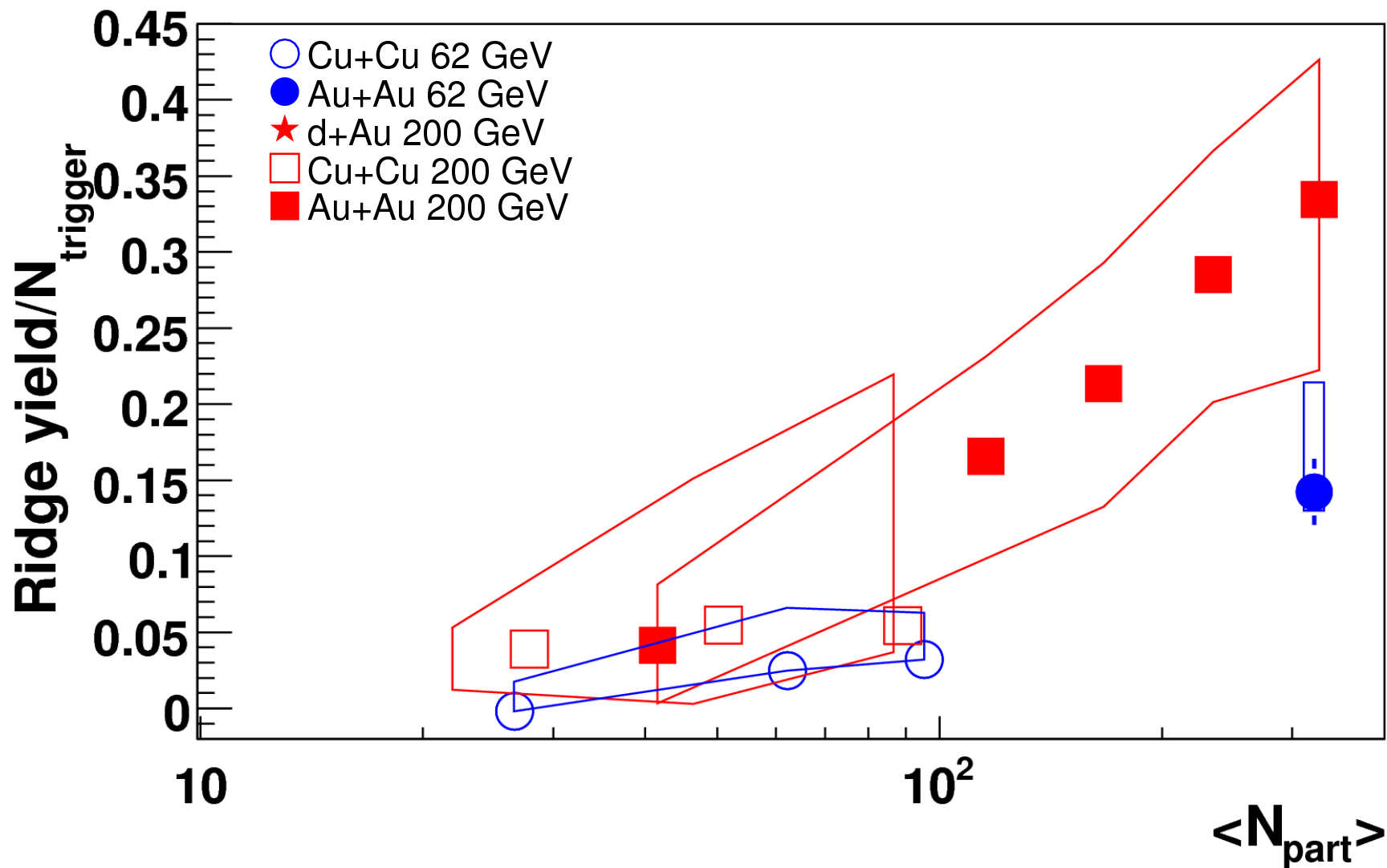


FIG. 7: Dependence of *Ridge* yield on N_{part} . The *Ridge* yield at $\sqrt{s_{NN}} = 62$ GeV is about 40% lower than at $\sqrt{s_{NN}} = 200$ GeV. Color online.

Figure 8

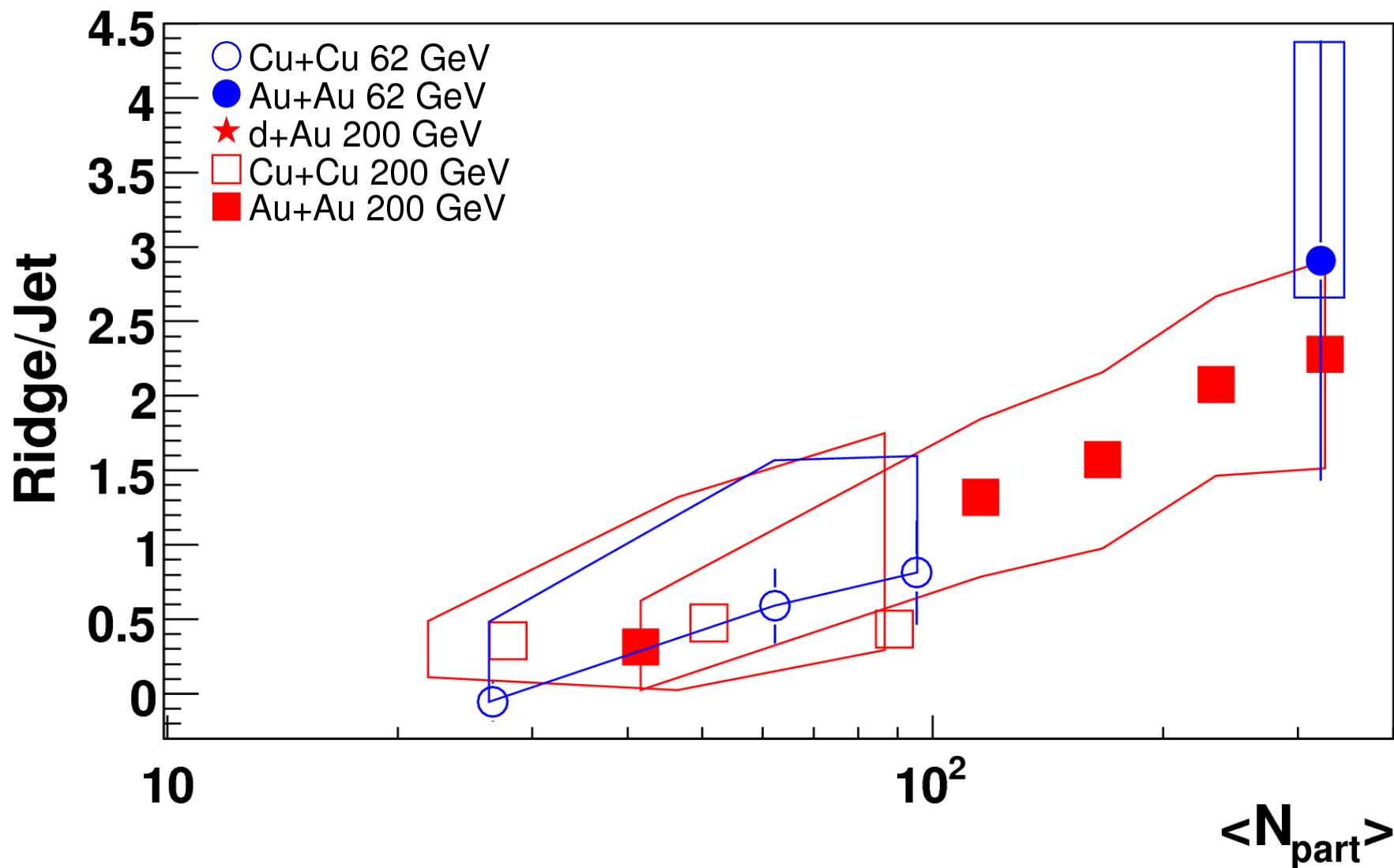


FIG. 8: Dependence of $Ridge/Jet$ on N_{part} . This ratio is independent of energy within errors, at least in the kinematic range investigated. Color online.

Conclusions

- Jet
 - Mostly independent of system, including Cu+Cu, Au+Au, and d+Au
 - Mostly consistent with Pythia up to a scaling factor
 - Differences could be
 - Pythia doesn't work well at low p_T
 - Incomplete subtraction of Ridge
 - But still remarkable Pythia does so well for something in heavy ion collisions and that there is so little system dependence
 - Pythia simulations imply same trigger p_T selection at two different energies corresponds to different z_T , jet energy
- Ridge
 - Consistent (within large errors) at same N_{part} for different collision systems
 - Ridge/Jet independent of energy (within this kinematic region)

VI. TO-DO LIST

1. Implement track merging correction and run over data again. May or may not need to run over $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV again.
2. Touch up Pythia simulations.
3. After track merging corrections are done play with getting yield from $\Delta\eta$ instead, bin counting
4. Analysis note and plots of correlations which yield each data point
5. Need $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV data for figure 1 and figure 2 - currently have $Au + Au$ $\sqrt{s_{NN}} = 62$ GeV data as placeholder